THE EDUCATION UNIVERSITY OF HONG KONG

Course Outline

Part I

Programme Title	**:**	Bachelor of Education (Honours) (Secondary)
Programme QF Level	**:**	5
Course Title	**:**	Foundations of Information and Communication Technologies
Course Code	**:**	INT2065
Department	**:**	Mathematics and Information Technology
Credit Points	**:**	3
Contact Hours	**:**	39 hours
Pre-requisite(s)	**:**	Nil
Medium of Instruction	**:**	English
Course Level	**:**	2

Part II

The University's Graduate Attributes and seven Generic Intended Learning Outcomes (GILOs) represent the attributes of ideal EdUHK graduates and their expected qualities respectively. Learning outcomes work coherently at the University (GILOs), programme (Programme Intended Learning Outcomes) and course (Course Intended Learning Outcomes) levels to achieve the goal of nurturing students with important graduate attributes.

In gist, the Graduate Attributes for Sub-degree, Undergraduate, Taught Postgraduate, Professional Doctorate and Research Postgraduate students consist of the following three domains (i.e. in short "PEER & I"):

- **P**rofessional **E**xcellence;
- **E**thical **R**esponsibility; **&**
- **I**nnovation.

The descriptors under these three domains are different for the three groups of students in order to reflect the respective level of Graduate Attributes.

The seven GILOs are:

1. Problem Solving Skills
2. Critical Thinking Skills
3. Creative Thinking Skills

4a. Oral Communication Skills

4b. Written Communication Skills

5. Social Interaction Skills
6. Ethical Decision Making

7. Global Perspectives

1. Course Synopsis

This course provides students with fundamental knowledge and skills in information and communication technologies (ICTs) so that they are competent to study more advanced ICT-related subjects.

2. Course Intended Learning Outcomes ($CILO_s$)

Upon completion of this course, students will be able to:

$CILO_1$ state and explain different hardware and software concepts of a computer system, and data representation inside computer;

$CILO_2$ explain the concepts of modern networks and Internet;

$CILO_3$ acquire the knowledge and skills for different applications software; and

$CILO_4$ examine the impact of ICT in society.

3. Content, CILOs and Teaching & Learning Activities

Course Content	CILOs	Suggested Teaching & Learning Activities
Introduction to Computer Hardware and Software: • Understanding digital components; • The language of computers: bits and bytes; • Input devices, output devices, processing, storage, and connectivity; • Operating system fundamentals; • System software and application software; • Collaborative application software. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding computer hardware and software.	$CILO_1$	Lectures, workshop, hands-on practices
Principles of Data Communication and Network: • Data communication fundamentals; • Networking fundamentals; • Network architectures and components; • Client/server network and topologies;	$CILO_{1,3}$	Lectures, hands-on practices

• The Internet and its applications; • Digital devices and media. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding principles of data communication and network.		
Introduction to Software Programming: • Understanding the importance of software programming; • The life cycle of software development; • Exploring programming languages; • Basic HTML, JavaScript, and Cascading Style Sheets programming. • Using AI to facilitate the teaching/learning to deliver the subject knowledge regarding software programming.	*CILO*$_{2,3}$	Lectures, hands-on practices
Impact of IT and AI in Society: • Computer viruses; • Cybercrime and identity theft; • Cryptography; • Privacy and security: • copyright, and ethical issue; • Impact of information and communication technology; • Technology and innovation for the future of products. • Considering the critical and ethical concerns behind when engaging with AI to learn and teach these computer concepts, guided by Education Bureau (EDB) (2026)'s AI Literacy Learning Framework for Primary and Secondary Schools.	*CILO*$_{4}$	Lectures, case study analysis, on-line discussion

4. Assessment

Assessment Tasks	Weighting (%)	CILO
(a) Group Project: Propose and teach a topic/ subject knowledge that is mentioned in this course. A 10-minute presentation (plus 5-minute Q&A) of the work is required, which	50	*CILO*$_{2,3,4}$

should cover the relevant computer concepts, learning objectives, pedagogy, and lesson activities, as well as its ethical and social implications. Each group should consist of 4-6 students.		
(b) Reflection Essay: Building on their group project outcomes and, students will conduct and report (in 1,500 English words) a small-scale teaching reflections by *interviewing a relevant stakeholder* (e.g., in-service teacher, student, classmate) about the lesson design, as well as its ethical and social implications; students will synthesize interview insights to refine their design, and reflect on how the AI tools may support or hinder meaningful learner development.	50	$CILO_{1,2,3,4}$

5. Required Text(s)

Nil

6. Recommended Readings

Capron, H.L. (2004). *Computers: Tools for an information age* (8th ed.). NJ: Prentice Hall.

Daley, B. (2006). *Computer are your future 2006* (Complete ed.). N.J.: Prentice Hall

Education Bureau (EDB) (2026). AI Literacy Learning Framework for Primary and Secondary Schools. https://www.edb.gov.hk/en/edu-system/primary-secondary/applicable-to-primary-secondary/it-in-edu/debp.html

Evans, A., Martin, K., & Poatsy, M.. (2017). *Technology in Action, Introductory* (13th ed.). Harlow, United Kingdom: Pearson Education Limited.

Evans, A., Martin, K. & Poatsy, M. (2015). *Technology in Action Complete, Global Edition*. Harlow: Pearson Education Limited.

Ellie Quigley (2004). *JavaScript by example.* N.J.: Prentice Hall.

Gralla, P., et al. (2002). *How the Internet works: Millennium edition* (1st ed.). Que Corporation.

Hennessy, J. L., & Patterson, D. A. (2017). *Computer architecture: A quantitative approach* (6th ed.). San Francisco: Morgan Kaufmann.

Lemay, L., Colburn, R., & Kyrnin, J. (2016). *SAMS teach yourself HTML, CSS & JavaScript web publishing in one hour a day* (7th ed.). Ind.: Pearson Education.

Pfaffenberger, B., Daley, B., & Baber, R. (2004). *Computers in your future 2004, Complete* (6th ed.). NJ: Prentice Hall.

Schafer, S. (2005). *HTML, CSS, Javascript, Perl, and PHP programmer's reference.* US: Wrox.

Vermaat, M. E., Sebok, S. L., Freund, S. M., Campbell, J. T., & Frydenberg, M. (2015). *Discovering computers 2016.* Boston, Mass.: Thomson/Course Technology.

7. Related Web Resources

Basic Education Curriculum Guide – To Sustain, Deepen and Focus on Learning to Learn (Primary 1 – 6)
http://www.edb.gov.hk/en/curriculum-development/doc-reports/guide-basic-edu-curriculum/index.html
Blueprint for Digital Education Development in Primary and Secondary Schools
https://www.edb.gov.hk/en/DEBP/
Key Learning Area Curriculum Guides (Primary 1 - Secondary 3)
http://www.edb.gov.hk/en/curriculum-development/kla/technology-edu/curriculum-doc/index.html
Updating of the Information and Communication Technology Curriculum and Assessment Guide (Secondary 4-6)
https://www.edb.gov.hk/attachment/en/curriculum-development/renewal/CM/EDBCM21099E.pdf
PC Webopedia
http://www.pcwbopedia.com
The Big6 Skills
http://big6.com

8. Related Journals

Nil

9. Academic Honesty

The University upholds the principles of honesty in all areas of academic work. We expect our students to carry out all academic activities honestly and in good faith. Please refer to the *Policy on Academic Honesty, Responsibility and Integrity* (https://www.eduhk.hk/re/uploads/docs/00000000001633679892454 8BbN5). Students should familiarize themselves with the Policy.

10. Others

Nil

Last update: 18 March 2019